

Vighnesh Poojary

Vighneshpoojary49@gmail.com | 7619161634

Github.com/vighnesh1477 | linkedin.com/in/Vighnesh Poojary



SUMMARY

Results-driven Information Science and Engineering student and researcher passionate about Artificial Intelligence, Software Development, and solving real-world problems through innovative engineering. First-author IEEE author with experience in cyber security, image processing, and AI research, complemented by a published patent application and multiple research publications. Skilled at transforming research into practical, scalable solutions while building secure, high-impact products.

EDUCATION

Mangalore Institute of Technology & Engineering, Moodabidri	Jun 2023 – Present
B.E. in Information Science & Engineering	CGPA: 8.39/10
Sri Durgaparameshwari Temple Pre University College, Mandarthi (KSEEB)	2023
Senior Secondary (12th)	Percentage: 85.16%
Karnataka Public School, Kokkarne (KSEEB)	2021
Secondary School (10th)	Percentage: 81.28%

SKILLS

Languages: Java, Python, JavaScript, SQL

Technologies & Tools: ReactJS, HTML, CSS, JavaScript, Firebase (Auth, Database, Hosting), Supabase (SQL), MySQL, Git, GitHub, Vercel, VS Code, CodeBlocks, Kali Linux, MS Office, Overleaf

Relevant Coursework: Object Oriented Programming, Databases, Data Structures and Algorithms, Operating Systems, Embedded system and design, Computer Networks, Machine Learning, Full-stack Development, Image Processing, Open source and Technologies, Digital design and systems, frontend, Design and Analysis of Algorithms, Cryptography, Cloud Computing, Business Intelligence, System Engineering.

PROJECT WORKS

Major Project:

Vehicle-to-Vehicle Signaling with Blind-Spot Detection for Safe Overtaking (Ongoing) - Group of 4

- Developing an intelligent overtaking assistance system using computer vision and V2V communication for real-time vehicle detection, tracking, speed and distance estimation, and hazard analysis to generate safe overtaking decisions.
- Approaches & Technologies:** RT-DETR with enhanced vehicle tracking, one-way/two-way road classification, environmental detection (day/night/rain/fog) using classical computer vision, road geometry prediction (curvature, visibility, reliability), pothole detection using K-means clustering, and optical V2V communication.

Patent:

- EcoRide Bin: A Smart Waste Management for Cleaner Journey (2026):** Developed a smart waste management system for electric buses featuring dual-compartment waste collection, sensor-driven segregation, and smart monitoring for timely waste disposal. **Published Patent Application:** Indian Patent Office Journal, Issue 34/2026, 21 Aug. 2026. **Application No.:** 202541013074 A.

Research Publications:

- A Novel and Innovative URL Access Path Obscuring Algorithm for Enhancing Student Data Security in ERP Systems (2025):** Proposed a URL obfuscation algorithm that replaces predictable USNs with randomly generated codes

to protect student images and improve data privacy in ERP systems. Presented and published at IEEE INSPIRE 2025, Mangalore, India. **DOI:** 10.1109/INSPIRE67328.2025.11300636.

- **Dynamic Subtitle Color Adaptation Using K-Means Clustering (2026):** Developed a dynamic subtitle rendering approach using K-Means clustering to analyze lower-frame background colors and select suitable text and outline colors for improved readability. Published in Indiana Journal of Multidisciplinary Research, 6(4), 208–215. **DOI:** 10.5281/zenodo.21602324.
- **Personalized Destination Tracking and Alerting Using QR Ticketing and GPS for Public Transport (2026):** Developed a QR-based destination tracking system using GPS to provide real-time passenger alerts near destinations and notify conductors about missed stops. Published in Indiana Journal of Multidisciplinary Research, 6(4), 488–493. **DOI:** 10.5281/zenodo.21623415.
- **An Ethical AI Framework for Identifying, Auditing and Mitigating Bias in AI Models (2025):** Proposed an open-source framework for bias detection and mitigation using fairness audits, proxy bias detection with Cramér's V, privacy risk evaluation, and synthetic data regeneration. Presented and published at IEEE ICTCT 2025, Bangalore, India. **DOI:** 10.1109/ICTCT69201.2025.00031.

Open Source Contribution:

- **TechPrep (2026):** Built and maintain an open-source web platform that enables learners to master technical concepts through 4,000+ interactive MCQs across 18+ domains, with detailed explanations and daily question updates. **Technologies:** React.js, JavaScript (JSON/JS Data Management), HTML5, CSS3, Vercel, Git, GitHub.

Side Project Works:

- **AI Diary Fullstack Website (2025):** Built a full-stack website allowing users to record daily routines via text/voice, auto-save by date, and secure login using Google authentication. **Technologies:** ReactJS, Firebase (Auth, Database, Hosting).
- **Lab-Colab Website (2025):** Created a full-stack platform where students can share text/files in the lab; all logged-in users can view and download the files. **Technologies:** ReactJS, Firebase, Supabase.

COURSES & WORKSHOPS

- **Infosys Springboard:** Completed certifications in Machine Learning, Python, C, R Programming, Java, Statistics, and Frontend Development, along with Introduction to Data Science, Introduction to Deep Learning, Introduction to Natural Language Processing (NLP), Computer Vision 101, and Introduction to Robotic Process Automation (RPA).
- **NPTEL:** Successfully completed Data Mining with Elite Certification (69%).
- **Generative AI Mastermind:** Workshop by Outskill (2026) & BE 10X: AI Tools and ChatGPT Workshop (2026).
- **Medical Imaging (MONAI):** Completed sessions on “Building Deep Learning Models” and “Real-World Case Studies” using the MONAI framework by Dr. Kumar Rajamani (Cropin Technologies).
- **Cybersecurity & AI:** “Reactive to Predictive: Next Gen AI-Enabled Cyber Threat Intelligence” by Dr. Gaurav Sharma (University of Sheffield) & “Privacy and Security in the Era of AI” by Dr. Sheela Siddappa.
- **Network & Mobility:** “Intelligent Architecture for 5G/6G Networks” (Nokia Networks) & “Decision Making in EV/Mobility” (TCS/IISc).
- **Deloitte Technology Job Simulation:** Completed practical tasks in coding and development (Sept 2025).
- **Hands-on Workshops:** “Arduino UNO” workshop at MITE (May 2025).

ACHIEVEMENTS & ACTIVITIES

- **Global Honorable Mention, PTC Competition:** Ranked in the Top 5 teams globally for designing an energy-efficient space.
- **IEEE & Springer Paper Presentations:** Presented research papers at Acharya Institute of Technology (IEEE), RVIM (Springer), and MITE (IEEE).
- **Publication:** Published Four conference papers in IEEE.